



1.13

Function Model Selection and Assumption Articulation

Student Notes and Guided Practice

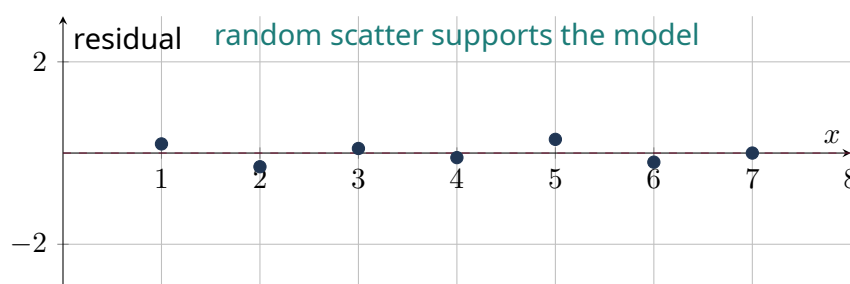
AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Select plausible linear, quadratic, exponential, or other models from data and context.
- Use residuals and structural evidence to compare models.
- State assumptions and limitations explicitly.

Essential Question

What evidence justifies choosing one model over another?



Concept Framework

Key Idea

Model choice should combine context, rate patterns, residuals, and domain.

Key Idea

Residual = observed - predicted. Random scatter around zero supports plausibility.

Key Idea

A strong fit does not prove causation or guarantee valid extrapolation.

Key Idea

Assumptions should address constancy of parameters and meaningful input range.

Formula Map

Formula and Structure

$$e_i = y_i - \hat{y}_i$$

Formula and Structure

$$\text{absolute error} = |e_i|$$

Formula and Structure

$$\text{relative error} = \frac{|e_i|}{|y_i|}$$

Worked Examples

Worked Example 1

Problem. A table has nearly constant ratios for equal input steps. Which model is most plausible?

Solution. An exponential model, provided the context supports proportional change.

Worked Example 2

Problem. A residual plot forms a clear U-shape. What does that say about a linear model?

Solution. The errors are systematic, so a linear model misses curvature and is not adequate.

Worked Example 3

Problem. Give one assumption behind an exponential population model.

Solution. For example, the proportional growth rate remains approximately constant and resources do not impose a carrying-capacity effect over the modeled interval.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is

incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Select or assess a model for the evidence: 2,5,8,11,14 at equal steps.

2. Select or assess a model for the evidence: 3,6,12,24,48 at equal steps.

3. Select or assess a model for the evidence: 1,4,9,16,25 at consecutive integers.

4. Select or assess a model for the evidence: residuals 1.2, -0.9, 0.7, -1.1, 0.8.

5. Select or assess a model for the evidence: residuals -4,-1,2,4,2,-1,-4.

6. Observed values are 20,31,48,70; model A predicts 21,30,49,69 and model B predicts 18,34,46,72. Compare MAE.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.

Video support: [Flipped Math topic page](#) | Original EHS material informed by Pearson and Holt Precalculus concepts.